# Chapter 11. Variable Selection

Consequences of omitting variables, search methods, selection criteria, Mallows' $C_p$

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With many candidate regressors we want a **parsimonious** model: enough variables to explain $y$ well and predict accurately, but no more, since every extra variable adds variance to the estimates and predictions. This chapter quantifies the trade-off and describes search methods and criteria.

## 11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where $X_k$ $(n \times (k+1))$ contains the intercept and $k$ regressors and $X_r$ $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

**Theorem 11.1**

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

**Proof.** (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var +bias $\cdot$ bias$'$.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var +bias$^2$ with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

**Theorem 11.2**

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

**Proof.** By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat\beta_k - \operatorname{Var}\tilde\beta_k = \sigma^2 AB^{-1}A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

In words: dropping variables whose coefficients are small relative to their standard errors (so that $\beta_r\beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces the mean squared error of estimation and prediction, even though it introduces bias. Conversely, if the reduced model is true ($\beta_r = \mathbf{0}$), both estimators are unbiased and the reduced one has smaller variance, so including irrelevant variables only costs precision.

## 11.2 Search methods

With $p$ candidate regressors there are $2^p$ subsets (each variable in or out), so exhaustive search becomes infeasible for large $p$ ($2^{50} \approx 10^{15}$).

**All possible regressions (best subset).** Fit all $2^p$ models. For each size $k$, keep the best model by $R_k^2$ (equivalently by $SSE_k$ or $MSE_k$, since these are monotone in each other for fixed $k$), then compare the best models across $k$ with a criterion that penalizes size (Section 11.3).

**Backward elimination.**

(1) Fit the full model.

(2) Compute the partial $F$ statistic of each variable in the model.

(3) Let $F_L$ be the smallest, and $F_C$ the critical value at level $\alpha_{\text{drop}}$.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

**Forward selection.** Start with no regressors. At each stage add the variable that increases $R^2$ the most (equivalently, has the largest partial $F$ given those already in), provided its partial $F$ exceeds the critical value at level $\alpha_{\text{add}}$; otherwise stop.

**Stepwise selection.** Forward selection, but after each addition the variables already in the model are re-examined by partial $F$ tests and removed if they fall below the $\alpha_{\text{drop}}$ threshold (with $\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling). A variable that was useful early can become redundant once others enter.

The methods need not agree, and none guarantees the best subset. The levels used are not true significance levels, because of the repeated testing.

**Example 11.1**

In Hald's classical cement data ($n = 13$, four ingredients $x_1, \ldots, x_4$), $x_2$ and $x_4$ are highly correlated ($x_1 + x_2 + x_3 + x_4 \approx 100$). Backward elimination ($\alpha = 0.05$) removes $x_3$ (partial $F = 0.02$) and then $x_4$ (partial $F = 1.86 < F_{0.05}(1, 9) = 5.12$), ending with $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters $x_4$ first (largest single $R^2 = 0.675$), then $x_1$, and stops because $x_2$ has partial $F = 5.03 < 5.12$, ending with $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise selection with $\alpha_{\text{add}} = 0.10$ enters $x_2$ and then removes $x_4$ (partial $F = 1.86$), ending with $\{x_1, x_2\}$. Because $x_2$ and $x_4$ carry nearly the same information, different routes end at different, almost equally

good models.

## 11.3 Selection criteria

For a model with $k$ regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R_k^2 = 1 - SSE_k/SST$;

(3) adjusted $R^2$: $R_{a,k}^2 = 1 - \dfrac{n-1}{n-k-1}(1-R_k^2) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the $MSE$ of the full model.

$R_k^2$ never decreases when a variable is added, so it can only compare models of the same size. $MSE_k$ can increase when a variable is added (the loss of one degree of freedom may outweigh the decrease in $SSE$), and $R_{a,k}^2$ is a decreasing function of $MSE_k$: maximizing $R_a^2$ is minimizing $MSE_k$.

**Mallows' $C_p$.** We judge the reduced model by the total standardized mean squared error of its fitted values at the $n$ design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}_{ki}'\tilde\beta_k.$$

**Theorem 11.3**

$$\Gamma_k = (k+1) + \frac{\beta_r' X_r'(I-H_k)X_r\beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence $C_k$ estimates $\Gamma_k$, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

**Proof.** By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}_{ki}'(X_k'X_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X_k'X_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}_{ki}'\Big\} = \mathrm{tr}\{(X_k'X_k)^{-1}X_k'X_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta_r'X_r'(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ∎

So we look for models with small $C_k$ and $C_k$ close to $k+1$; a $C_k$ well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.